



Guide: capacitive touch sensor

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Github repo for code example: <https://github.com/DCC-Lab/IntelligentCage/blob/master/CapacitiveSensor/CapacitiveSensorSketchExample/CapacitiveSensorSketchExample.ino>

What is it ? A capacitive touch sensor can sense proximity of a conductor. It uses electrical capacitance as an indicator of presence of a conductor, like a finger or other body parts. The sensor is simply a metallic plate, foil, tip or any other metallic object. It acts as a half-capacitor while the finger acts as the other grounded half. Without any finger close to it, the metal piece creates a capacitor with infinity/the closest conductor. Since capacity drops as the distance between the two sides get further from each other, the capacitance is small. On the other hand, when a finger is close or touching the metal plate, the capacitance is high. Using this property, it is possible to detect the touch or proximity of someone or an animal, which acts as a grounded plate.

Circuit implementation To use this type of sensor with Arduino, two digital pins are required. The first one (pin 3 on figure 1) is the sending pin (output) and the second one (pin 4 on figure 1) is the receiving pin (input).

A RC circuit is used to determine the capacitance of the sensing capacitor built by the metal plate and the potential presence of a body. A RC circuit is composed a resistor (R) and capacitor (C). The following equation describes the charging of the capacitor

$$V_C(t) = V_0(1 - e^{-\frac{t}{RC}}) \quad (1)$$

The RC parameter is called the time-constant τ and is linked to the charging time. A high capacitance (or resistance) will increase the time required to charge the capacitor. This charging time is a great parameter to measure and link it to capacity, which is linked to the presence of a body.

To measure this charging time, the sending digital pin gets from LOW to HIGH and acts as a constant voltage source. At first, since the capacitor is empty of charges, the current will flow to ground according to Ohm's law. Then, while the capacitor begins to charge itself, the current will decrease.

Finally, we will reach a point where the capacitor is full. A full capacitor acts as an "open-circuit" and the current drops to zero. According to Ohm's law, the voltage dissipated by the resistor is zero since there is no more current. That means that the voltage is now the same at the sending pin and between the resistor and

the capacitor. This is only true when the capacitor is full or, in other words, when the current is zero.

So, by measuring the delay between the moment the send pin gets HIGH and the moment the voltage between the resistor and the capacitor is the same, we have the charging time of the capacitor. After that, it can easily be linked to the capacity and finally detect the presence of a body.

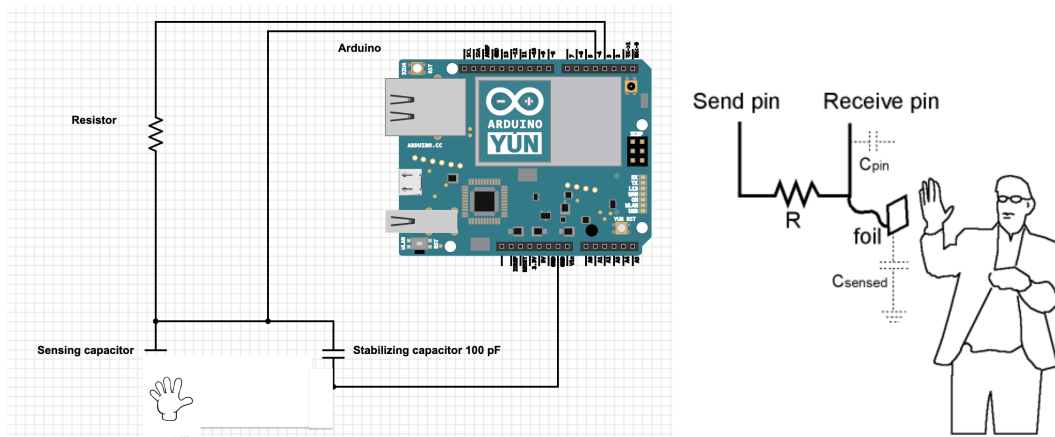


Figure 1: Circuit to implement with an arduino (left) and a more visual circuit showing the half metallic plate (right)(source: [here](#))

On figure ??, the sensing capacitor is the half

Arduino code The CapacitiveSensor Arduino library can be used to implement this type of sensor.

Github: <https://github.com/PaulStoffregen/CapacitiveSensor>.

Documentation: <https://playground.arduino.cc/Main/CapacitiveSensor/>

Touch or proximity? It is possible to change the sensitivity of the capacitive sensor by changing the resistor. It can be tweaked to get a touch-only sensor or a sensor that can sense at distance. The highest the resistance, the highest the sensitivity will be.

For a touch-only sensor the resistance should be between 1 M Ω and 3 M Ω .

For a distance sensing, the resistance should be over 10 M Ω .

It is always possible to manually calibrate the resistance and test a few combinations to get what is wanted in terms of sensitivity. It will also depend on what body is used. I didn't test with a mouse. I only did with my body that is not the same size as a mouse (of course). There is more free charges in my body than in a mouse so conductivity is higher. This will probably have an impact on the sensitivity of the sensor.

Stabilization of the reading To get a more stable reading of the charging time, a 100 pF capacitor can be added between the ground and the sensing pin. This capacitor

will compensate small discharge of the sensing capacitor by discharging itself when the sending pin voltage is oscillating (not a perfect constant 5 V).